Unit: Unit 0: Foundations of Algebra
Topic: Inequalities

Name: ____________
Date: ________

Advanced (Habanero)

Q1. Solve for x: $2x + 5 > 11$
(A) $x > 3$
(B) $x < 3$
(C) $x > -3$
(D) $x < -3$
(E) $x = 3$

Q2. Graph the solution to $x - 2 < 5$ on a number line
(A) $(-\infty, 7)$
(B) $(-\infty, 3)$
(C) $(-3, \infty)$
(D) $(-7, 3)$
(E) $(-7, \infty)$

Q3. Write the solution in interval notation: $x > -4$ and $x < 1$
(A) $(-4, 1)$
(B) $(-\infty, -4) \cup (1, \infty)$
(C) $(-4, \infty)$
(D) $(-\infty, -4) \cup (-4, 1)$
(E) $(-\infty, 1)$

Q4. Solve the inequality: $-3x + 2 < 5$
(A) $x > -1$
(B) $x < -1$
(C) $x > 1$
(D) $x < 1$
(E) $x = -1$

Q5. Solve for x: $x/2 + 3 > 7$
(A) $x > 8$
(B) $x < 8$
(C) $x > -8$
(D) $x < -8$
(E) $x = 8$

Q6. Graph the solution to $2x + 1 > 5$ on a number line
(A) $(-\infty, 2)$
(B) $(-\infty, -2)$
(C) $(2, \infty)$
(D) $(-2, \infty)$
(E) $(-2, 2)$

Q7. Write the solution in interval notation: $x < -2$ or $x > 3$
(A) $(-\infty, -2) \cup (3, \infty)$
(B) $(-2, 3)$
(C) $(-\infty, 3)$
(D) $(-2, \infty)$
(E) $(-\infty, -2) \cap (3, \infty)$

Q8. Solve the inequality: $x - 4 > 2x - 5$
(A) $x < 1$
(B) $x > 1$
(C) $x = 1$
(D) $x < -1$
(E) $x > -1$

Open Ended Questions (FRQ)

Q9. Solve for x and graph the solution on a number line: $3x - 2 > 2x + 5$

Q10. Write the solution in interval notation: $x + 1 > 4$ and $x - 2 < 1$

Chef's Tips

- Emphasize the importance of isolating the variable when solving inequalities
- Remind students to consider the direction of the inequality when graphing solutions on a number line
- Encourage students to use interval notation to represent solutions